MILITARY SPECIFICATION

VOLTMETER, MILITARIZED, SOLID STATE, DC
DIFFERENTIAL, GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-V-28712(EC), dated 23 April 1969, and is approved for use by all Departments and Agencies of the Department of Defense.

Page 1

2.1, SPECIFICATIONS, MILITARY: Add "MIL-C-28777/1-Cable Assembly, Electric Test Equipment, (3 Wire, 3 Prong, 125 Volt, Molded) Grounding Plug Connector, and MIL-C-5015 Connector with Cable Clamp, 60 Hertz".

Page 2

2.1, DRAWING, MILITARY: Delete "RED2689314 - Power Cable Assembly for Use with Electronic Test Equipment".

Page 4

3.6.2, line 2: Delete "and 230 volts".

3.6.2, line 3: Delete "±5 percent" and substitute "±3 Hz".

3.6.4: Delete "type A of Drawing RED2689314" and substitute "MIL-C-28777/1".

Page 5

3.7.1: Delete last sentence.

3.7.4: Delete and substitute:

"3.7.4 Enclosure. The complete equipment, with cover in place, shall be submergible (watertight for a 3-foot head in accordance with MIL-STD-108. The equipment, with cover removed, shall be dripproof in accordance with MIL-STD-108."

Page 13

4.7.2.2.3, (a): Delete "58.2" and substitute "58".

4.7.2.2.3, (b), line 1: Delete "56.4" and substitute "56".

4.7.2.2.3, (b), line 2: Delete "58.2" and substitute "58".

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TABLE III: Delete and substitute:

" TABLE III. Frequencies to be used for frequency transient tests.

Starting frequency(Hz)	Transient frequency(Hz)
62	64
48	46
51	53
367	356
432	444

"

Page 16

4.7.3.4: Delete.

Page 18

4.7.3.7.1.1: Delete.

4.7.3.8: Delete and substitute:

"4.7.3.8 Enclosure test. The following enclosure tests shall be performed after the shock and vibration tests."

Add new paragraph:

"4.7.3.8.1 Dripproof (cover removed) test.

(a) Perform the satisfactory operation tests of 4.7.3.2.

(b) Mount the equipment, deenergized and in its combination case with the cover removed, on a platform or tilt table in a manner such that the equipment can be oriented horizontally and in any of four directions - forward, backward, and to each side - to the extent of 15 degrees from the horizontal plane.

(c) Attach a spray nozzle with a uniformly distributed full cone spray pattern to a flexible hose and position the nozzle 3 feet above the equipment so that the spray is directed downward onto the equipment.

(d) Using a container calibrated in gallons or an equivalent method, determine the rate of flow out of the nozzle. The requirements for water flow shall be 4.33 gallons (1000 cubic inches) per hour for each square foot of area covered by the spray 3 feet below the nozzle. The equipment shall be subjected to a uniform spray over the entire surface being tested.

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(e) During the test, the unit shall be subjected to the following conditions:

(1) Spray directed onto the top (with platform or tilt table horizontal) for at least 8 minutes. Spray turned off 4 minutes.

(2) Spray directed downward onto the equipment for at least 8 minutes with the platform or table tilted such that the equipment is oriented 15 degrees from the horizontal in the forward position. Spray turned off 4 minutes.

(3) Spray directed downward onto the equipment for at least 8 minutes with the platform or table tilted such that the equipment is oriented 15 degrees from the horizontal in the backward position. Spray turned off 4 minutes.

(4) Spray directed for at least 8 minutes onto each side with the platform or table tilted such that the side being sprayed is oriented 15 degrees.

(5) If the equipment has two specified operating positions, such as resting on the bottom or back, the bottom of the unit will be considered a side and shall be tested accordingly. It is not necessary to completely retest the unit in the second position, only retest with the bottom treated as a side.

(f) Return the equipment to the horizontal (platform or tilt table horizontal) and repeat the satisfactory operation test of 4.7.3.2. Failure of the equipment to operate satisfactorily will constitute failure of this test. Remove the instrument from the case and note the presence of water. Any accumulation of water that presents a possible hazard to the equipment or to the operator of the equipment will constitute failure of this test.

"4.7.3.8.2 Submersion test (cover installed).

(a) Perform the satisfactory operation tests of 4.7.3.2.

(b) Attach the cover to the combination case. Remove and replace the cover three times.

(c) Perform the tests for submergible (watertight) equipment in accordance with MIL-STD-108, to a depth of three feet.

(d) Repeat the satisfactory operation tests of 4.7.3.2. Failure of the equipment to operate satisfactorily will constitute failure of this test. Remove the instrument from the case and note the presence of water. Any accumulation of water that presents a possible hazard

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to the equipment or to the operator of the equipment will constitute failure of this test."

4.7.3.9: Delete.

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4.7.4.5.1: Delete.

5.1: Delete and substitute:

"5.1 Packaging. Packaging shall be in accordance with MIL-P-116, method III."

5.1.1: Delete.

5.1.1.1: Delete.

5.2.1, last sentence: Delete and substitute: "Loose fill polystyrene shall not be used."

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5.4: Delete.

Preparing activity:
NAVY - EC
(Project 6625-N553)

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